Machine Learning PROJECT REPORT

HEART DISEASE PREDICTION

Submitted by

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ABSTRACT

Heart disease is one of the main causes of death in the world. Accurate and timely prediction of heart disease can improve patient outcomes by enabling timely diagnosis and preventive measures. This project develops a machine learning-based model to determine the risk of heart disease based on a variety of health factors, such as age, blood pressure, cholesterol, BMI, and lifestyle choices. The dataset is preprocessed and analysed using the Python libraries Pandas, NumPy, and Scikit-learn. The Logistic Regression algorithm is used to separate the patients into two groups: "Heart Disease" and "No Heart Disease." With a 90% training accuracy and an 87% testing accuracy, the model showed excellent predictive performance. The results demonstrate the effectiveness of machine learning in helping medical professionals assess a patient's risk and guide treatment choices.

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1. INTRODUCTION

One main use of machine learning in healthcare is predicting heart disease. Traditional diagnosis techniques often rely on manual interpretation of test results and clinical experience. This can lead to delays and incorrect diagnoses. In contrast, machine learning models can analyze large amounts of medical data, reveal hidden patterns, and make accurate predictions.

The goal of this project is to create a predictive model that uses various medical characteristics to identify individuals at high risk for heart disease. The dataset for this project includes features like blood pressure, cholesterol, triglyceride levels, and other important physiological parameters for assessing cardiovascular health. By automating the prediction process, the proposed system reduces human error and enables effective early detection.

The objectives of this project are to:

- Develop a reliable machine learning model to forecast heart disease.
- Preprocess and clean the dataset to ensure accuracy and consistency.
- Evaluate the performance of the Logistic Regression model using a range of metrics.
- Examine model generalisation by contrasting testing and training accuracy.

2. IDENTIFYING THE PROBLEM

Heart disease is one of the most deadly illnesses in the world and a leading cause of premature death. Determining who is at risk is challenging due to the complex interactions between a number of factors, such as lifestyle, genetics, and medical history. Traditional diagnostic methods usually depend on medical testing and expert opinion, which can lead to missed or delayed diagnoses. Many people don't realise they're at risk until a serious event, such as a heart attack, has happened, which increases treatment costs and decreases recovery chances.

Manually analysing vast amounts of patient data is time-consuming and error-prone. It goes without saying that we need a system that can automatically learn from historical health data, spot trends, and generate accurate and useful predictions regarding heart disease risk. This project aims to bridge that gap by developing a predictive system that can help physicians and patients quickly identify heart disease risks.

3. OBJECTIVES OF THE THESIS

The main goal of this project is to create a reliable machine learning system that can predict a patient's risk of heart disease based on their medical and demographic information. This system will help healthcare professionals make better and faster decisions.

Specific objectives include:

1. Identify and Prepare the Data: To get the data ready for modelling, analyse it, deal with missing values, remove noise, and normalise it.

2. Identify Key Factors: Look at the traits that have the greatest impact on the risk of heart disease, such as age, sex, blood pressure, blood sugar, and cholesterol.

3. Make use of models for machine learning: Compare and contrast different algorithms, such as Random Forest, K-Nearest Neighbour (KNN), Support Vector Machine (SVM), Decision Tree, and Logistic Regression.

4. Evaluate Model Performance: Use accuracy, precision, recall, and F1-score to gauge model performance in order to determine the best algorithm.

5. Create a Predictive System: Create a system that can identify if a patient is at risk for heart disease based on the input health parameters.

6. Promote Preventive Healthcare: Provide patients and doctors with the tools they need to take preventative measures and make better healthcare decisions.

4. Data Collection and Preprocessing

The Heart Disease Prediction System was developed methodically using machine learning methodology. Data collection, data preprocessing, model construction, model assessment, and system deployment are among the phases. Every stage will guarantee that the system can predict heart disease with the highest level of accuracy, dependability, and capacity.

1. Information gathering

The study's dataset was collected from publicly accessible heart disease datasets and medical databases like the UCI Machine Learning Repository. The dataset was selected because it contains a variety of medical parameters that will be used as input features when the model is run during the training and testing stages. These parameters include age, gender, blood pressure, cholesterol, fasting blood sugar levels, resting valvular ECG results, and maximum heart rate.

2. Preprocessing data

Missing values, inconsistent data, and superfluous features are frequently found in raw data. Prior to running the model, data preprocessing is an essential step in ensuring the quality of the data. The actions that will occur are as follows:

Managing missing values: Replace or remove null (or missing) values as needed.

Data normalisation: To make distorted readings consistent, the majority of numerical data should be scaled to a common range.

The process of turning categorical data into numerical data is known as "encoding categorical variables."

Finding the pertinent characteristics involved in an incident involving heart disease is known as feature selection.

3. Data division

A training dataset and a testing dataset are created from the preprocessed dataset. While the testing dataset refers to the model, the training dataset is used to train the model.

5. Methodology

1. Dataset description

The dataset is comprised of patient records that have numerous medical attributes in which to predict the chance of heart disease.
Some examples of features are:

Age

Gender

Blood Pressure

Cholesterol Level

Exercise Habits

Smoking

Family Heart Disease

BMI

Alcohol Consumption

Stress Level

Sleep Hours

Sugar Consumption

Triglyceride Level

The target variable is:

1 – Heart disease

0 – No heart disease

2. Data preprocessing

The preprocessing step allows the data to be cleaned and properly prepared to train: the dataset had missing values were either imputed or removed; all numeric values had been normalized to the similar range; categorical.

3. Train-Test Split

The cleaned dataset was divided into:

- **Training Set:** 80% of data
- **Testing Set:** 20% of data

  This ensures that the model is trained on a large portion of data and tested on unseen data to evaluate its generalization capability.

6. Implementation

1. Tools and Libraries Used

The Google Colab environment was used to implement the project in Python. The libraries listed below were utilised:
• Pandas & NumPy: For numerical computation and data manipulation.
• Matplotlib and Seaborn: For correlation analysis and data visualisation.
• Scikit-learn: For putting the Logistic Regression model and assessment metrics into practice.

2. Model Building

The Logistic Regression algorithm was selected for this classification problem because it provides a simple yet effective method for binary prediction tasks. It models the probability that a given input belongs to a particular class using a logistic function. The model was trained using the preprocessed features (X_train) and matching labels (y_train). The target variable was the degree of heart disease (1 or 0). The logistic regression model can be expressed mathematically as follows:

$$P(y = 1|x) = 1/\left(1 + e^{-(\beta^0+\beta^1 x^1+\beta^2 x^2+\cdots+\beta_n x_n)}\right)$$

3. Model Evaluation

The model that was trained was assessed using:

- Accuracy Score
- Confusion Matrix
- Classification Report (Precision, Recall, F1-Score)
- ROC-AUC Curve
- Cross-validation

To identify overfitting or underfitting, the accuracy of the training and testing processes was also compared.

Python Notebook:

https://colab.research.google.com/drive/1U0G2WdPwnu9ezfwLNfr8IbxCZ8J-B2HP?usp=sharing

7. Results and Discussion

The confusion matrix demonstrated that most of the predictions matched the actual results, and the ROC-AUC score was above 0.85, indicating strong discriminative power.

The small difference in accuracy between training and testing shows that the model has good generalisation on unknown data. Because of this, the Logistic Regression model is suitable for real-world applications where reliability and interpretability are crucial.

A graphical representation of the findings was also created using Seaborn and Matplotlib in order to better understand feature correlations and the distribution of disease status across various parameters.

8. Conclusion and Future work

This project successfully developed a machine learning-based model for heart disease prediction using logistic regression. The model did well and demonstrated that it could successfully classify patients based on their health characteristics, with a test accuracy of roughly 87%.

Important conclusions include:

- Preprocessing the data was essential for increasing accuracy;
- Logistic regression was dependable and interpretable;
- The model had no discernible overfitting, guaranteeing strong generalisation.

Future Enhancements:

• Add additional health indicators like ECG readings, smoking habits, and stress levels.
• To increase accuracy, try out more complex models like Random Forest, XGBoost, or Neural Networks.

• Create a real-time prediction application for the web or mobile devices.

• For intuitive interaction, use Flask or Streamlit to deploy the model.

REFERENCES

- Scikit-learn Documentation: https://scikit-learn.org/stable/
- Pandas Library Documentation: https://pandas.pydata.org
- Kaggle- heart_disease dataset

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